

Methods: NGC 4429

ALMA Thesis Planner (automated pipeline)

1. Data preparation and inspection

The student begins by loading the two delivered FITS products — the HCN(1–0) spectral-line cube (320×320 pixels, 126 channels, ~14.65 MHz / ~49.6 km/s channel width) and the Band 3 continuum image (combining spw 17/21/23/25, centered near 86.4/88.3/98.4 GHz) — into a Python analysis environment using ‘astropy.io.fits’, ‘spectral-cube’, and ‘radio-beam’ (CASA’s ‘imstat’, ‘immoments’, and ‘imview’ tasks are an acceptable alternative for any step). The student confirms the WCS, pixel scale, rest frequency (88.6316 GHz for HCN(1–0)), and beam parameters (recorded directly from the FITS headers: $\approx 1.88'' \times 1.81''$ at PA $\sim 80^\circ$ for the cube, $\approx 1.63'' \times 1.55''$ at PA $\sim 80^\circ$ for the continuum) by printing the header and cross-checking against the values quoted in the proposal metadata. The student explicitly notes in a lab notebook that these measured beams are coarser than the archive’s “representative” $0.094''$ figure, and that all subsequent physical-scale conversions use the measured, not representative, beam.

2. Feasibility check: noise characterization

Before any scientific measurement, the student characterizes the noise properties of the HCN(1–0) cube. Using line-free channels (identified by eye from a quick-look spectrum extracted at the nuclear pixel and from a few off-nuclear pixels within the primary beam), the student computes the per-channel RMS with ‘spectral-cube’s statistics tools or CASA ‘imstat’, restricting to a region within the $\sim 60''$ primary beam to avoid primary-beam-correction noise inflation at the map edges. This produces a noise curve across the 126 channels, which the student inspects for channels affected by atmospheric lines, edge-channel roll-off, or artifacts. This RMS curve is the basis for later detection-significance and upper-limit calculations, and its purpose here is solely to confirm that HCN(1–0) is plausibly detectable (or usefully constrainable) in this dataset — no detection is assumed in advance.

3. Moment map construction

Using the line-free channels to define and subtract a local baseline, the student constructs moment 0 (integrated intensity), moment 1 (intensity-weighted velocity), and moment 2 (velocity dispersion) maps of the HCN(1–0) cube with ‘spectral-cube’s ‘moment()’ methods (or CASA ‘immoments’), masking on a signal-to-noise threshold (e.g., $>3\sigma$ per channel, using the RMS curve from Step 2) to avoid noise bias in the moment-0 map. Maps are produced both for the full $\sim 60''$ field of view and for a zoomed nuclear postage stamp ($\sim 10''$ across, ~ 20 – 30 beams centered on the SMBH) to examine the circumnuclear region at the relevant scale. All maps are displayed with the synthesized beam ellipse overlaid for reference.

4. Radial intensity profile

The student defines the pixel position corresponding to RA 12:27:26.503, Dec +11:06:27.58 (converted to pixel coordinates via the FITS WCS) as the nuclear center and extracts an azimuthally

averaged radial profile of the moment-0 (integrated HCN intensity) map in concentric annuli of width equal to roughly half the beam FWHM, out to several beams' radius, using 'photutils' or a custom annulus-averaging routine. The student converts angular radius to projected physical radius (in pc) using one literature-standard distance to NGC 4429 (e.g., from a standard distance catalog such as NED or a Virgo Cluster distance compilation), explicitly citing the chosen value and treating it as an adopted external input rather than a pipeline-derived quantity. The student then inspects the profile for a central deficit, plateau, or monotonic peak, and reports the result strictly as a beam-resolution-limited test: since the previously reported CO(2–1) hole ($\sim 10\text{--}100$ pc) is comparable to or smaller than the $\sim 1.8''$ beam (which itself corresponds to a physical scale computed from the same adopted distance), the thesis states clearly that a flat or peaked profile within one beam cannot distinguish a genuine sub-beam cavity from a filled but beam-smoothed nucleus, and does not claim to re-measure the true size or depth of the hole.

5. Upper-limit calculation (in the event of a non-detection)

If no statistically significant nuclear HCN(1–0) emission is found (per the masking threshold in Step 3), the student computes a rigorous upper limit at the nuclear position using the per-channel RMS from Step 2, integrated over a velocity range matching the galaxy's known systemic velocity and a plausible nuclear line width (a few resolution elements, chosen conservatively and stated explicitly), following the standard radio-astronomy upper-limit formula (e.g., $3\sigma \times \text{channel width} \times \sqrt{N_{\text{channels}}}$ within a synthesized beam area). This flux limit is converted to a beam-averaged HCN column density and then to a dense-gas mass upper limit using one literature-standard HCN-to-H₂ conversion (α_{HCN}) factor, explicitly cited and flagged as an assumption rather than a measured quantity. The student repeats this conversion across a small, bounded set (e.g., 3–4 values spanning the commonly cited range) of alternative literature α_{HCN} values found in the dense-gas literature, and reports the resulting mass limit as a range rather than a single number, to make the sensitivity to this assumption transparent.

6. Continuum morphology characterization

Independently, the student examines the Band 3 continuum image at and around the SMBH position. Using 'imstat'/'photutils', the student measures the local RMS noise in continuum-image regions away from the nucleus, then checks for a statistically significant compact source coincident with the nuclear coordinates versus extended or diffuse emission, fitting a 2D Gaussian (e.g., with 'imfit' in CASA or 'astropy.modeling') to any detected compact component and comparing its deconvolved size to the measured beam ($\approx 1.63'' \times 1.55''$) to assess whether it is resolved or point-like. The student reports position, peak flux density, and integrated flux density (with uncertainties from the map RMS) for any detected nuclear component, or a flux upper limit if none is detected, using the same RMS-based method as in Step 5. The student explicitly states, as a caveat carried through to the discussion, that a single narrow-band ($\sim 86\text{--}98$ GHz) continuum measurement cannot alone distinguish AGN synchrotron/free-free emission from thermal dust or stellar continuum, so this measurement is used only as a morphological and positional constraint, not as an AGN diagnostic.

7. Synthesis

Finally, the student combines the results of Steps 3–6 into a single discussion table/figure showing: (a) the nuclear HCN(1–0) moment-0 map and radial profile (detection or upper limit, with the explicit beam-resolution caveat from Step 4); (b) the dense-gas mass limit or measurement range from Step 5; and (c) the continuum morphology result from Step 6. The student uses these three lines of evidence jointly to characterize which of two qualitative scenarios the data are more

consistent with — a genuinely gas-poor cavity (low/undetected HCN plus no notable compact nuclear continuum) versus an unresolved or tracer-dependent hole (HCN plausibly present but below the current beam resolution or sensitivity) — tying the discussion back to the proposal’s stated AGN-feedback-versus-secular-origin question. The student explicitly notes that a definitive answer would require the CO(1–0) and CO(6–5) observations mentioned in the original proposal, which are not present among this member OUS’s delivered products and are therefore outside the scope of the analysis performed.